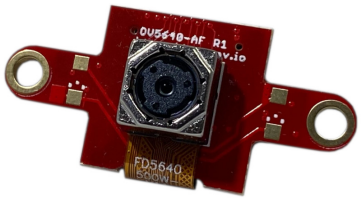
OpenMV® OV5640 FPC Camera Module

Datasheet

SKU: OMV-MOD-OV5640-V1

Description

The OV5640 FPC Camera Module gives you the same high-resolution JPEG-taking capabilities as the OpenMV Cam H7 Plus and RT1062's OV5640 camera module in a low-profile form factor with autofocus support. The OV5640 image sensor can capture 2592x1944 (5 MP) images. The sensor supports Grayscale, RGB565, JPEG, YUV422, and Bayer image formats. Depending on the resolution, you can achieve up to 120 FPS frame rates.



Target Areas

High-resolution imaging, JPEG capture, autofocus applications, low-profile installations, barcode scanning, document capture, quality inspection.

Key Features

- **5 MP resolution** — 2592x1944 pixel capture with the OV5640 sensor.
- **Autofocus** — Built-in autofocus support.
- **Multiple formats** — Grayscale, RGB565, JPEG, YUV422, and Bayer output.
- **Up to 120 FPS** — Frame rate depends on resolution.
- **Low-profile FPC** — Flexible printed circuit form factor for thin installations.
- **Universal compatibility** — Works with all modular OpenMV Cam base boards.

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1 Specifications

1.1 Sensor Specifications

Component	Details
Sensor	OmniVision OV5640 CMOS image sensor.
Resolution	2592x1944 (5 MP).
Autofocus	Yes — built-in autofocus support.
Output Formats	Grayscale, RGB565, JPEG, YUV422, Bayer.
Max Frame Rate	Up to 120 FPS (resolution-dependent).
Form Factor	Low-profile flexible printed circuit (FPC).

1.2 Compatibility

Component	Details
Supported Cameras	All modular OpenMV Cam base boards.

2 General Info

Dimension	Value
Weight	10 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/ov5640-fpc-camera-module
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.